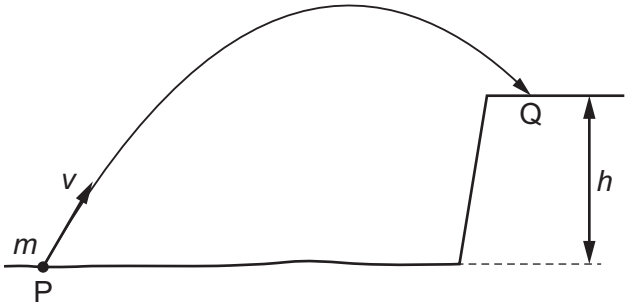
9 The diagram shows an object of mass m being projected with speed v from a point P to a point Q on top of a cliff height h.

The object does work W against air resistance as it travels from P to Q.



What is the kinetic energy of the object when it reaches point Q?

A $\frac{1}{2}mv^2 + mgh + W$

B $\frac{1}{2}mv^2 + mgh - W$

C $\frac{1}{2}mv^2 - mgh + W$

D $\frac{1}{2}mv^2 - mgh - W$